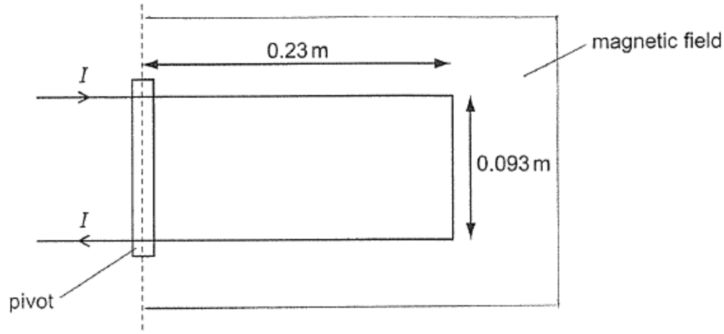
22. A current, *I* of magnitude 9.6 mA is passed into a current balance which consists of a U-shaped wire of negligible mass placed in a region of constant magnetic field which is in the plane of the paper and perpendicular to the pivot. The U-shaped wire has length 0.23 m and the arms are 0.093 m apart, as shown in the diagram below.



The U-shaped wire experiences a turning moment about the pivot of value 4.7×10^{-6} N m

What is the magnitude of the magnetic flux density of the constant magnetic field?

A 5.27 mT **B** 22.9 mT **C** 45.8 mT **D** 4.37 T

Ans: B

Magnetic force experienced by the wire, $F_B = BIL\sin\theta$

Moment produced, $F_B \times d = 4.7\times10^{-6} \Rightarrow (BIL\sin\theta)\times d = 4.7\times10^{-6}$

Therefore, magnetic flux density,

$$B = \frac{4.7\times10^{-6}}{(IL\sin\theta)d} = \frac{4.7\times10^{-6}}{(9.6\times10^{-3})(0.093)(\sin 90^\circ)(0.23)} = 22.9\,\text{mT}$$

23. Four identical wires A, B, C and D carry currents of equal magnitude, in the